RS Flavor Constraints

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1 Combined Constraints

The minimum value of Y_{5D} (which Yukawa the constraint refers to depends on the type of particle, see next sections) vs. KK mode mass can be seen in Fig. 1. Assuming a common M_{KK} scale for all sectors, the minimum Yukawa for each sector can be identified as the upper edge of the constrained region for that value of M_{KK}. The value of g_{s*}, the strong coupling constant, can vary, see below Eq. 2.19 in [1]. For no brane kinetic terms, [1] claims $g_{s*} \approx 6$ at tree level throughout the region of interest, while [2] uses $g_{s*} \approx 3$ at one loop level. Also following [1, 3, 4] etc., we indicate the Yukawa value were the system becomes strongly coupled for a number of perturbative KK modes (N_{KK}), or

$$Y_{5D} < \frac{2\pi}{N_{KK}} \tag{1}$$

on most plots. This prevents constraints from being alleviated by simply going to arbitrarily high Yukawa.

2 Quarks: Down Sector

The leading constraints are from the $\Delta F = 2$ operators.

2.1 $\Delta F = 2$ constraints

The current generic $\Delta F = 2$ constraints on new physics are provided by the UTFit collaboration in [5] (see [6] for more detailed information from the collaboration. The leading constraints are from the operator $Q_4^{q_i q_j} = \bar{q}^{\alpha}_{jR} q^{\alpha}_{iL} \bar{q}^{\beta}_{jL} q^{\beta}_{iR}$ in Kaon oscillation. UTFit defines the coefficients $C_i(\Lambda)$ as

$$C_i(\Lambda) = \frac{F_i L_i}{\Lambda^2} \tag{2}$$

where F_i are functions of NP flavor couplings and L_i are loop factors. Assuming $F_i \sim L_i \sim 1$, we obtain

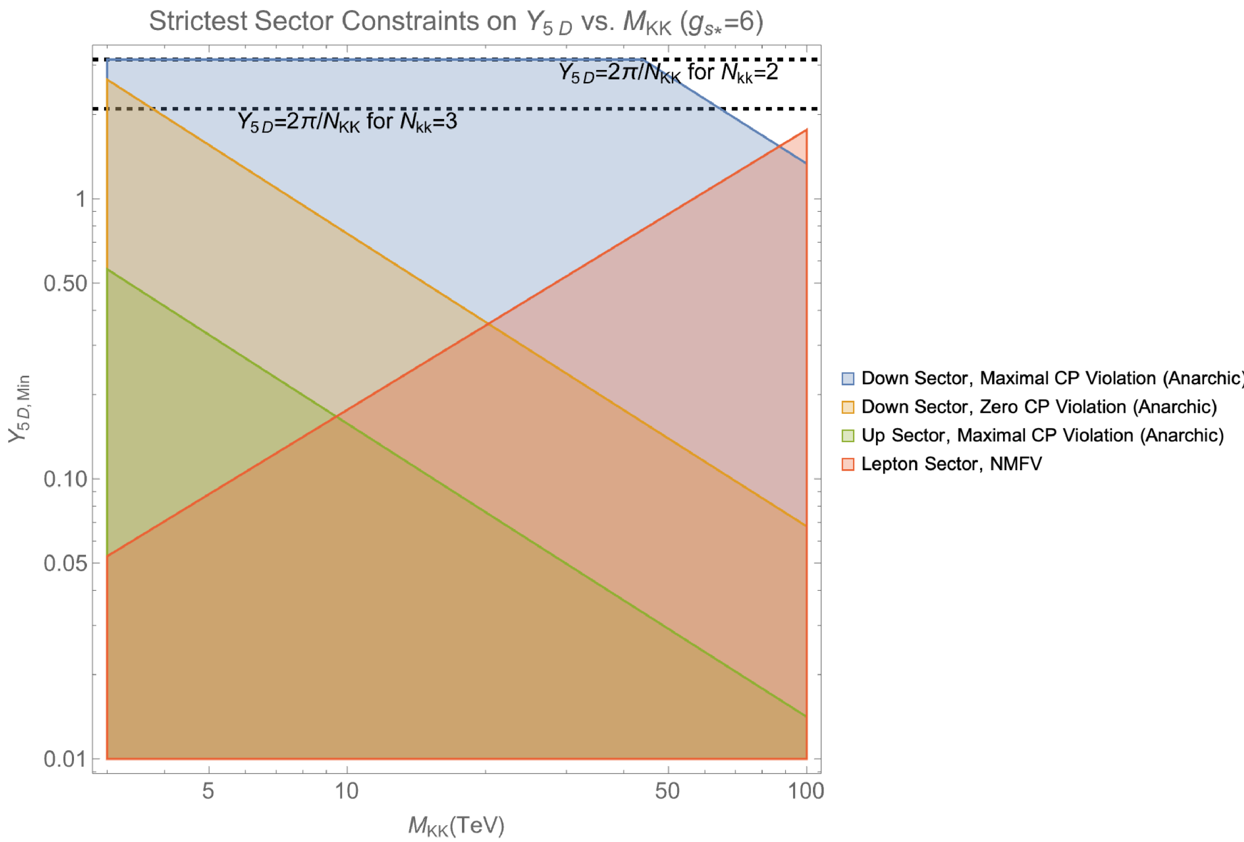
$$\begin{aligned} \Lambda_{\text{NP}} &\approx 2.45 \times 10^4 \text{ TeV (No CP violation)} \\ \Lambda_{\text{NP}} &\approx 4.83 \times 10^5 \text{ TeV (Maximal CP violation)} \end{aligned} \tag{3}$$

where the no CP violation constraints come from constraints on $\text{Re}(C_{4K})$ and maximal CP violation constraints from $\text{Im}(C_{4K})$.

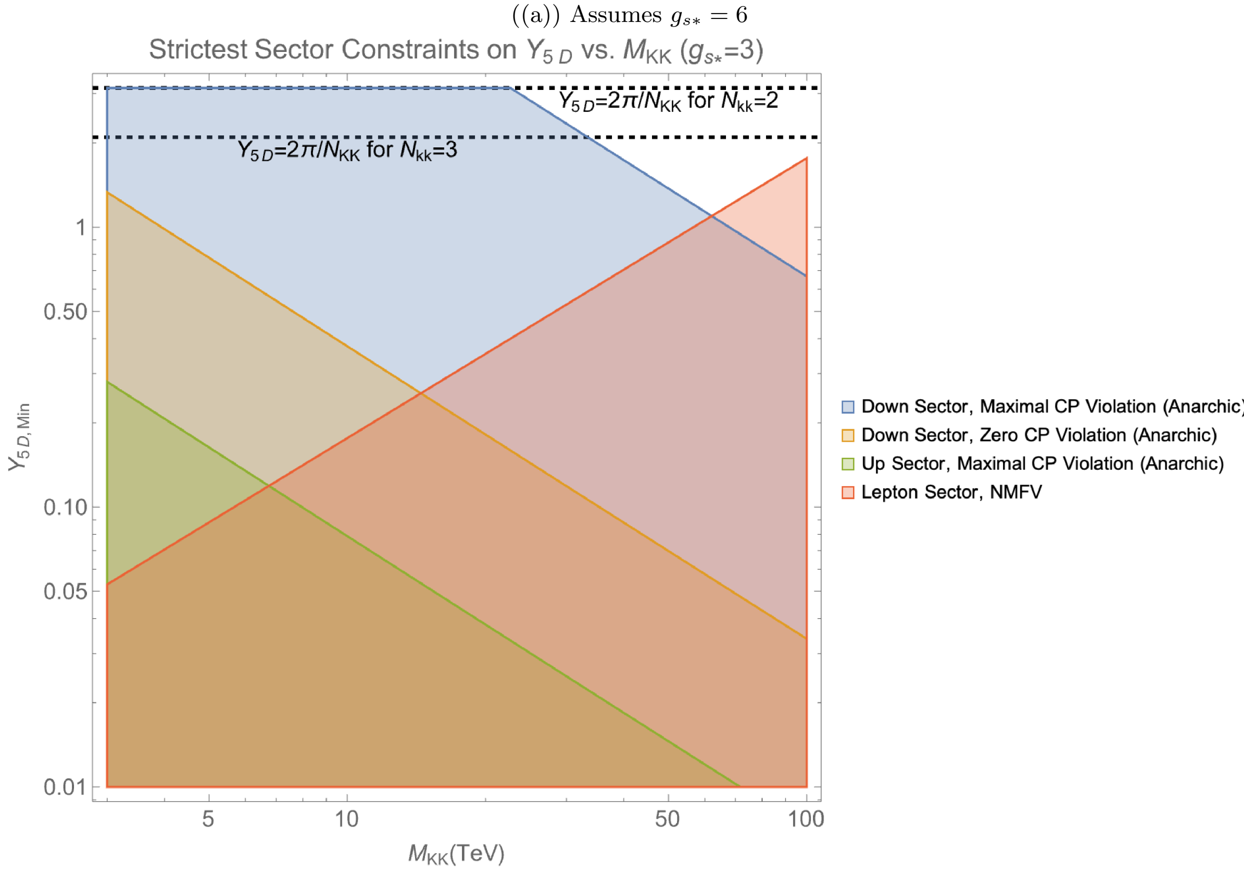
2.1.1 Anarchic RS Constraints

In RS with anarchic Yukawa matrices [3],

$$|C_{4K}| \sim \frac{g_{s*}^2}{M_G^2} \frac{1}{Y_{d,5D}^2} \frac{2 m_d m_s}{v^2}, \tag{4}$$



((a)) Assumes $g_{s*} = 6$



((b)) Assumes $g_{s*} = 3$

Figure 1: The strictest constraints on down quarks, up quarks, and leptons that I have identified. See later sections for information on exactly what Yukawa the figure refers to in a given sector.

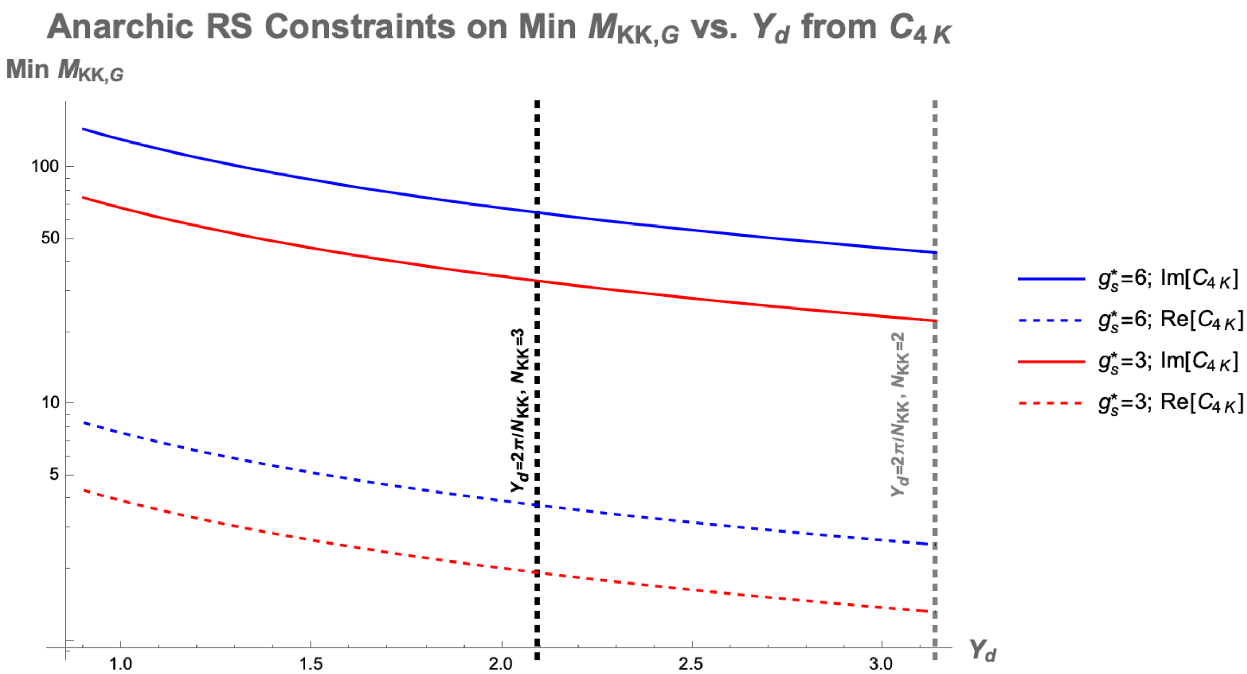


Figure 2: Plot of the minimum value of M_G for a given value of $Y_{d,5D}$ (Eq. 5) to meet the constraints from Eq. 3. $Y_{d,5D}$ values are at the M_G scale in question.

where g_{s*} is the 5D strong coupling and M_G is the KK gluon mass. We then recover the minimum value of M_G for a given value of $Y_{d,5D}$ by requiring

$$|C_{4K}| \sim \frac{g_{s*}^2}{M_G^2} \frac{1}{Y_{d,5D}^2} \frac{2m_d m_s}{v^2} \sim \frac{1}{\Lambda_{\rm NP}^2} \tag{5}$$

for the $\Lambda_{\rm NP}$ values in Eq. 3 after running the values to M_G. The constraints for different values of g_{s*}[1] can be seen in Fig. 2 (for values of Y_d at the scale in question). Note that stricter constraints can be obtained if one assumes that NP physics only contributes a percentage C_{4K}.

2.1.2 NMFV

For the reasons expressed in [3], numerical evaluation is required to accurately determine the suppression, hence it is not shown here.

3 Quark Sector: Up Quarks

Note that there are some questions related to the SM predictions in $D_0 - \bar{D}_0$ mixing, see [7].

3.1 $\Delta F = 2$ constraints

Again, the leading constraints are from the $Q_4^{q_i q_j}$ operator. From [5], we find

$$\Lambda_{\rm NP} \approx 4.77 \times 10^4 \text{ TeV (Maximal CP violation).} \tag{6}$$

[5] does not give updated constraints for minimal CP violation, but based on the older values in [2] it is likely an order of magnitude smaller.

[1] See [1] for reasons why that value might change

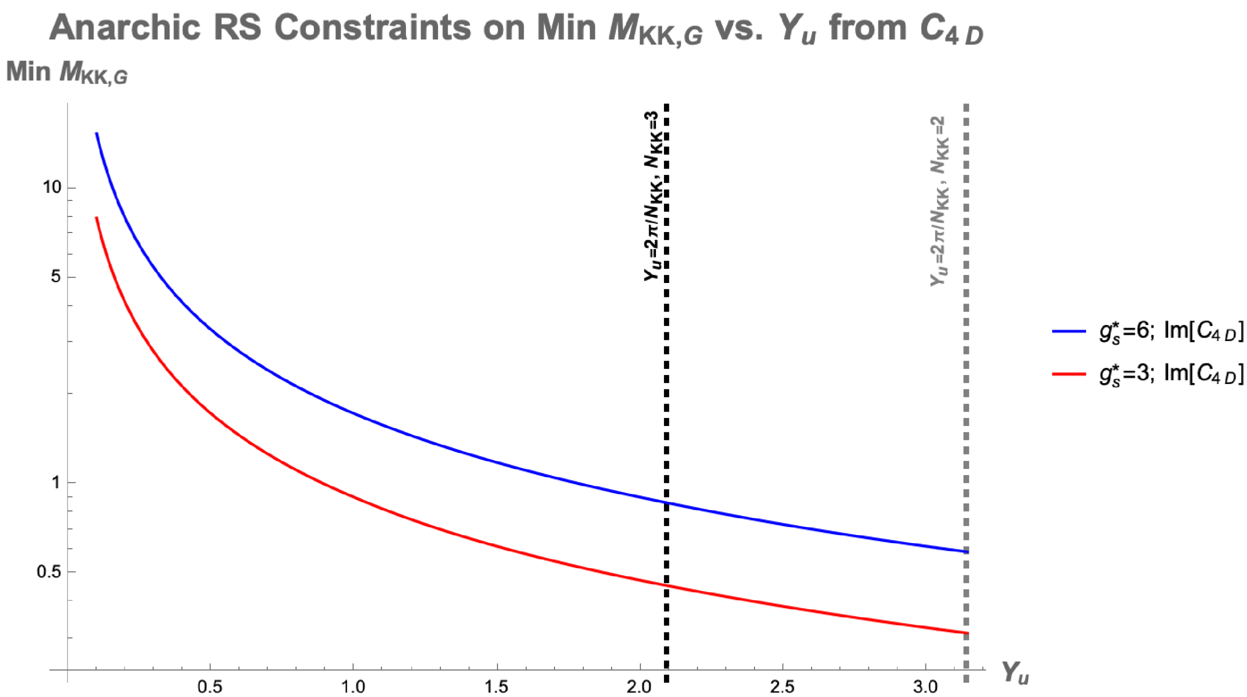


Figure 3: Plot of the minimum value of M_G for a given value of $Y_{u,5D}$ (Eq. 7) to meet the constraints from Eq. 6. $Y_{u,5D}$ values are at the M_G scale in question.

3.1.1 Anarchic RS Constraints

In RS with anarchic Yukawa matrices, we obtain [2]

$$|C_{4D}| \sim \frac{g_{s*}^2}{M_G^2}\frac{1}{Y_{u,5D}^2}\frac{2m_c m_u}{v^2}\gamma(c_{Q_2})\gamma(c_{u_2}) \sim \frac{1}{\Lambda_{\text{NP}}^2}, \tag{7}$$

where $\gamma(c)$ is a correction due to the quark overlap with the first KK gluon mode (see [1], $\gamma(c_{u_2}) \sim \gamma(c_{Q_2}) \sim 1.2$). Following [2] in using $c_{u_2} = 0.53$ and $c_{Q_2} = 0.58$, the constraints on the minimum value of M_G can be seen in Fig. 3. Note that we have again assumed all of the value of C_{4D} is produced by NP.

4 nEDM

For anarchic nEDM, nEDM measurements could be problematic (see [8]), but less so than C_{4K}. However, for NMFV, because the only source of CP violation is the CKM matrix as in the SM, we do not expect contributions to the nEDM before 3 loops. Therefore, SM contributions will still dominate.

5 Leptons

The strongest flavor constraint in the lepton sector is $\mu \to e\gamma$ constraint.

5.1 $\mu \to e\gamma$

[9] lists the current constraint on $\mu \to e\gamma$ BR as

$$\text{BR}(\mu \to e\gamma) < 3.1 \times 10^{-13}. \tag{8}$$

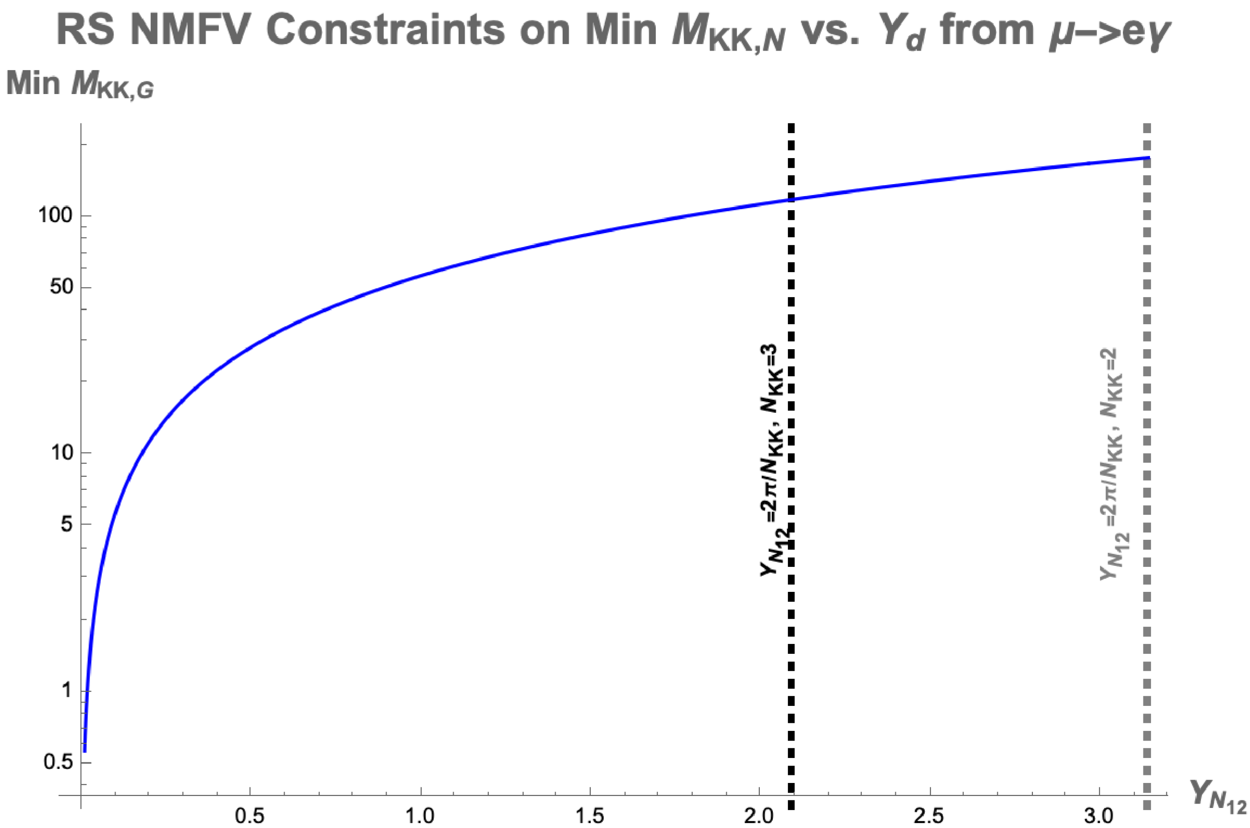


Figure 4: Plot of the minimum value of $M_{N,KK}$ for a given value of $Y_{N_{12}}$ (Eq. 9) to meet the constraints from Eq. 8.

5.1.1 RS NMFV

Under the NMFV assumptions, [4] finds

$$M_{KK,N} \sim (\frac{3e^2}{\text{BR}(\mu \to e\gamma) G_F^2 4\pi^2})^{1/4} \frac{|\bar{Y}_{N_{12}}|}{2}, \tag{9}$$

where $\bar{Y}_N$ corresponds to the same mass conventions as in [3] (keep in mind [3] and [4] use different Higgs mass conventions), G_F is the 4 fermion interaction constant etc. Without running any of the couplings, for the given branching ratio, this corresponds to the linear relation

$$M_{KK,N} \sim 56.7\bar{Y}_N \text{ TeV}. \tag{10}$$

The reason that I consider this constraint as less constraining on the overall KK mode scale is that [4] seems to expect $\bar{Y}_N \sim \mathcal{O}(0.1)$, not order one like for the quark yukawas in [3]. A plot can be seen in Fig. 4.

References

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